

Homopolar Bearingless Slice Motor With Flux-Biasing Halbach Arrays

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Abstract—In this article, we present a new configuration of bearingless slice motor that levitates and rotates a ring-shaped solid steel reluctance rotor. The rotor is 50 mm in diameter and has salient features on the outer surface. Symmetric sets of Halbach magnet arrays, mounted on the tips of stator teeth, establish a homopolar bias flux around the rotor. The bias flux passively stabilizes the rotor in the out-of-plane tilts and axial translation, whereas the rotor's radial translations are actively stabilized by feedback control. The rotor saliencies spatially modulate the air-gap bias flux such that the resulting torque-current relationship is similar to that of permanent-magnet synchronous machines. We have designed, built, and tested a prototype bearingless motor and control system. The prototype system achieves a torque constant of 14.9 mNm/A, maximum speed of 5500 r/min, and suspension bandwidth of 84 Hz with a phase margin of 11.3°. The rated torque and speed are 26.8 mNm and 3486 r/min, respectively, and the axial and tilting passive stiffnesses are 15.3 N/mm and 34.4 mNm/°, respectively.

Index Terms—Bearingless motors, flux biasing, homopolar machines, slice motors.

I. INTRODUCTION

BEARINGLESS motors are special types of electric machines where the rotor is levitated and rotated by a single stator unit. Not only a driving torque, but also constraint forces to center the rotor, are magnetically transmitted from the stator without mechanical contact, thereby eliminating the need for mechanical bearings [1], [2]. One of the challenges in bearingless motors comes from the need for feedback stabilization of unstable axes, which is typically realized with extra hardware components, e.g., sensors, windings, and power electronics. To mitigate such an issue, bearingless slice motors

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have been developed by using a permanent-magnet rotor whose length-to-diameter ratio is relatively small [3], [4]. Such a disk-shape geometry passively stabilizes the rotor's axial translation and out-of-plane tilts. Therefore, active stabilization is only needed for two radial translations. The compact form factor and contact-free operation make bearingless slice motors advantageous for some special applications, e.g., high-purity liquid pumps [5] and blood pumps [6].

Some configurations of bearingless slice motors have been developed over the last 20 years, but research efforts have been made only recently toward magnet-free rotors [7]. Motivations for such rotors arise from the drawbacks of permanent magnets, such as high material cost, brittleness, and demagnetization at high temperatures [8], [9]. Commonly used magnet-free rotors in bearingless slice motors are reluctance rotors using a magnetically soft material with saliencies [9]–[12], or hysteresis rotors using a magnetically semihard material with smooth surface [13]. These rotors are typically flux biased by permanent magnets embedded in the stator for passive stabilization. For example, the stators in [10] and [13] have an axially magnetized permanent magnet at the center, and the stator in [12] has two axially magnetized permanent-magnet rings and iron rings for homopolar flux biasing. On the other hand, the stators in [9] and [11] have multiple permanent magnets alternately embedded in the stator for multipole flux biasing.

In this article, we present a new bearingless slice motor that levitates and rotates a ring-shape reluctance rotor. Two sets of Halbach magnet arrays, mounted symmetrically on the tips of stator teeth, interact with the rotor's saliencies to establish a homopolar bias flux in the air gap. Our new flux-biasing scheme enables a single-sided magnetic design where bias flux and control flux are routed only through the rotor's outer surface. Also, the Halbach arrays steer the bias flux more toward the rotor, thereby reducing the leakage bias flux and improving the suspension passive stiffness. The bias flux is spatially modulated by the rotor saliencies such that the resulting flux distribution in the air gap contains an eight-pole component as well as a homopolar component. To interact with these components in a decoupled fashion, the stator is equipped with two sets of windings which individually generate eight-pole magnetomotive force (MMF) and two-pole MMF. This winding scheme enables decoupled control of rotation and suspension. The design and preliminary test result of our bearingless motor have been presented in [14]. This article further discusses the winding design, estimation and control algorithms, and performance evaluation via finite-element analysis (FEA) and various hardware tests.

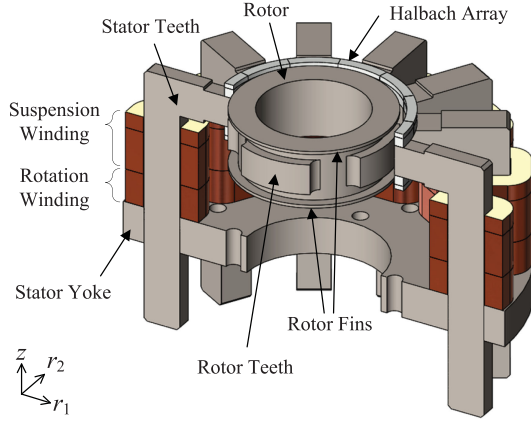


Fig. 1. Magnetic design of our bearingless slice motor. The stator is sectioned to provide an internal view.

The rest of the article is organized as follows. Section II illustrates the magnetic design and explains the operating principle. Section III presents a prototype system and discusses the design and construction. Section IV explains the control and its implementation. Section V presents the test results and Section VI gives conclusions and suggestions for future work.

II. OPERATING PRINCIPLES

The magnetic design of our bearingless motor is shown in Fig. 1. The stator core consists of a yoke and 12 inverted L-shaped teeth forming a bore, in which a ring-shaped reluctance rotor is inserted. The rotor has three layers of saliencies on the surface: two radial fins at the top and bottom and four rotor teeth in the middle. There are two annular magnet arrays mounted at the top and bottom of the tips of the stator teeth, aligned with the rotor's radial fins. The stator teeth are wound with multiple coils comprising two sets of windings, i.e., suspension windings and rotation windings. The rest of the section explains the operating principles of the rotor suspension and rotation.

A. Suspension

Fig. 2 shows the flux-biasing scheme and rotor suspension principle. Two sets of magnet arrays are placed on the top and bottom of the stator teeth. Note that the magnetization pattern (small black arrows) is axisymmetric about the z -axis, where the magnets closer to the stator teeth are axially magnetized and the magnets farther from the stator teeth are radially magnetized. Such a magnet configuration, first proposed by Halbach [15], has been used in electric machines to effectively orient permanent-magnet flux toward a single side [16], [17]. In our design, two Halbach arrays push the permanent-magnet flux (blue dashed line) radially inward, which is then routed by the rotor saliencies toward the stator teeth. As a result, the air gap between the rotor teeth and stator teeth is biased with radially outward homopolar flux.

The bias flux generates reluctance forces that align the rotor with the stator teeth. This effect passively stabilizes the rotor's translation along the z -axis as well as tilts about the r_1 -axis

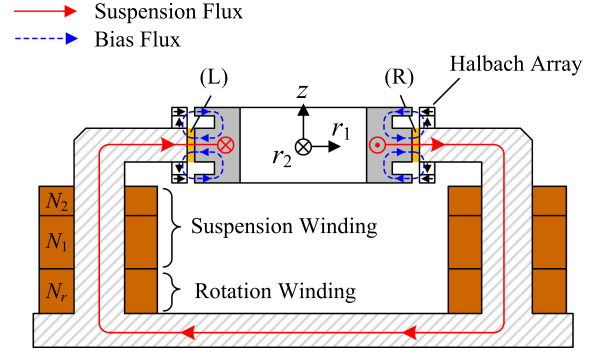


Fig. 2. Illustration for flux-biasing scheme and suspension force generation. The Halbach arrays at the top and bottom of the stator teeth establish the bias flux, and the suspension winding superimposes a control flux. As a result, the net flux density around the rotor strengthens in (R) and weakens in (L), thereby generating a reluctance force to the right to control the radial degrees of freedom.

and r_2 -axis. However, the bias flux also destabilizes the rotor's translations along the r_1 -axis and r_2 -axis. We actively control these radial translations by superposing a two-pole suspension flux (solid red line) on the bias flux, as shown in Fig. 2. Note that the suspension flux strengthens the net flux density in the area (R), whereas weakens it in the area (L), thereby generating a net flux-steered force toward the r_1 -axis. Active stabilization of the two radial translations can be implemented with a feedback controller that adjusts the relative angle and magnitude of the suspension flux based on the rotor position measurements. Due to the homopolar bias flux, the required magnitude of the two-pole suspension flux is zero when the rotor is centered. This allows relatively low copper loss from the suspension winding and low hysteresis and eddy-current drags on the rotor, which is a common characteristic of homopolar magnetic bearings [2].

B. Rotation

Fig. 3 schematically shows a top view of the rotor and stator, sectioned across the r_1r_2 -plane. Here, the rotor is displaced by an angle ϕ with respect to the r_1 -axis. Note that the air-gap bias flux (blue dashed lines) is more concentrated on the rotor teeth than the rotor slots. This air-gap flux distribution can be conceptually decomposed into homopolar plus eight-pole components. A driving torque can be generated when the stator applies an eight-pole rotating MMF in synchronism with the eight-pole component of the bias flux.

In principle, the resultant eight-pole flux does not interfere with the two-pole suspension flux to generate radial disturbance forces. This can be explained by the $p \pm 1$ pole-pair principle of bearingless motors [1, p. 274]; that is, the condition for p_1 pole-pair flux distribution to generate a radial force is to interact with flux distribution whose pole-pair number is

$$p_2 = p_1 \pm 1. \quad (1)$$

In our bearingless motor, the rotation flux has pole-pair number $p_r = 4$, and thus does not satisfy (1) with either the suspension flux ($p_s = 1$) or the homopolar component of the bias flux ($p_b = 0$). This property enables us to implement decoupled control for rotation and suspension, as discussed in Section IV in

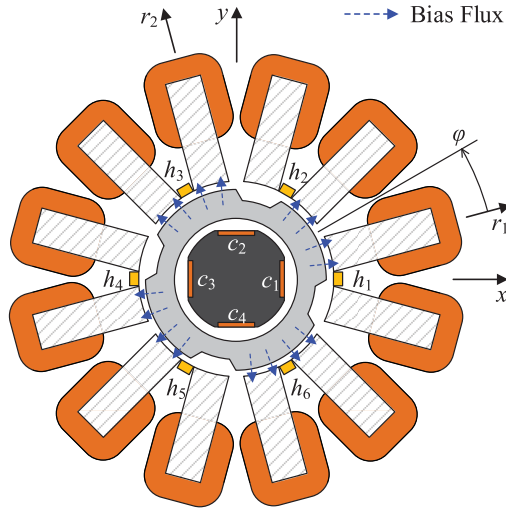


Fig. 3. Schematic top view of the machine. The bias flux is concentrated on the rotor teeth and displaced by an angle ϕ relative to the stator. The stator is equipped with four sensing coils (c_1 – c_4) for suspension control and six Hall sensors (h_1 – h_6) for rotation control.

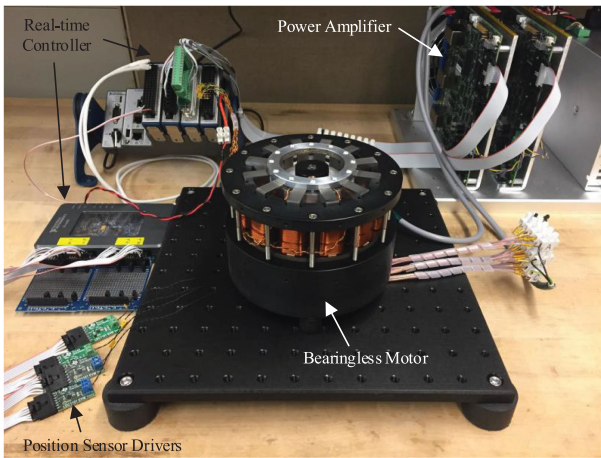


Fig. 4. Prototype bearingless motor and control system. The reluctance rotor is removed in this picture to show the inside of the stator bore.

details. Such a decoupling property in homopolar-type bearingless motors has been first studied by Ichikawa *et al.* [18].

III. DESIGN AND CONSTRUCTION

We have built a prototype system as shown in Fig. 4. This section explains the design and construction details of the stator, rotor, and windings.

A. Stator

Fig. 5(a) shows a section view of the stator. At the center, there is an aluminum reference structure against which the stator core and magnet arrays are assembled. The stator core is made of 0.35-mm thick lamination steel (M330-35A). Key design parameters of the stator are listed in Table I.

Fig. 5(b) and (c) shows the top and bottom magnet array assemblies, each consisting of an aluminum housing that

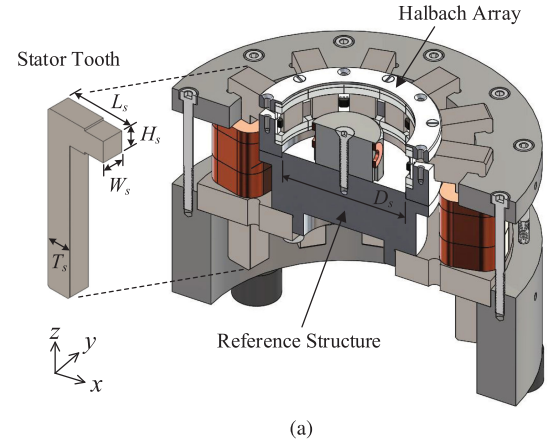


Fig. 5. Stator design and photographs. (a) Section view of the stator. (b) Top magnet assembly. (c) Bottom magnet assembly. The top magnet assembly is upside down to show the magnets.

TABLE I
STATOR DESIGN PARAMETERS

Symbol	Quantity	Value
L_s	Stator tooth length	57 mm
H_s	Stator tooth height	10 mm
W_s	Stator tooth width	10 mm
T_s	Stator tooth thickness	12 mm
D_s	Stator bore diameter	54 mm
g_o	Magnetic air gap	2 mm

supports an Halbach array. The bottom assembly has its housing castellated to have the stator teeth inserted. The Halbach arrays are built on the aluminum housings by stacking arc-shape NdFeB (N35) magnets with an adhesive (Loctite326 bonder and Loctite7649 primer). The arc-shape magnets have 1.5-mm thickness, 3-mm width, 27-mm inner radius, 30° arc angle, and two types of magnetization: One type is axially magnetized and the other type is magnetized radially inward. Each of the Halbach arrays consists of 36 magnets stacked in three layers. The magnets in the first two layers from the stator teeth are axially magnetized, whereas the magnets in the last layer are radially magnetized.

B. Rotor

Our reluctance rotor is shown in Fig. 6. Here, the photograph in Fig. 6(a) shows the actual rotor made of low-carbon steel (AISI 1018), and the computer aided design (CAD) model in Fig. 6(b) shows key geometric parameters.

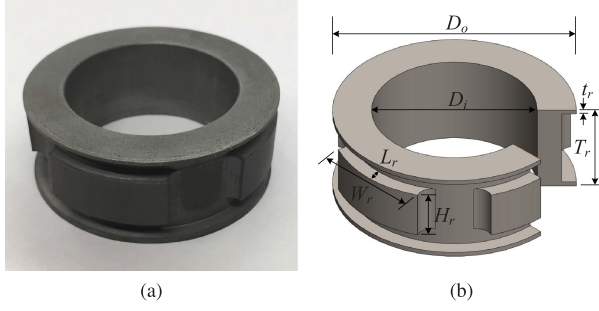


Fig. 6. Reluctance rotor (a) photograph and (b) CAD model showing the design parameters.

TABLE II
ROTOR DESIGN PARAMETERS

Symbol	Quantity	Value
D_o	Rotor outer diameter	50 mm
D_i	Rotor inner diameter	34 mm
L_r	Rotor tooth length	3 mm
H_r	Rotor tooth height	10 mm
W_r	Rotor tooth width	23 mm
T_r	Rotor overall thickness	19 mm
t_r	Rotor fin thickness	1 mm

The rotor saliencies are designed for effective magnetic coupling with the stator. There are two circular fins protruding radially outward, which aligns with the Halbach arrays' radially magnetized magnets. Between the two fins, there are four rotor teeth evenly displaced along the circumference. The rotor teeth are designed for the width W_r that minimizes cogging torque. We have performed three-dimensional (3-D) magnetostatic FEA simulations, using ANSYS Maxwell, and the results predict that the cogging torque is minimized when the rotor teeth width are approximately integer multiples of the stator teeth pitch, e.g., 11 or 23 mm. The actual rotor is designed for $W_r = 23$ mm. The detailed design parameters of the rotor are listed in Table II.

We have also computed the rotor's passive stiffness using FEA simulations. One figure of merit for bearingless slice motors, where achieving weak negative radial stiffness k_r and strong positive axial stiffness k_z are both important, is the absolute value of the ratio between the two passive stiffnesses, i.e., $|k_r/k_z|$ [19]. The lower this number, the easier the take-off and stabilizing control become for the same axial suspension stiffness. The theoretical minimum is $|k_r/k_z| = 0.5$, which is achievable if the magnetic bearing is composed of permanent magnets only [20]. Our design achieves $|k_r/k_z| = 1.7$. Although higher than the theoretical minimum, our value is ten times and three times lower than those of prior-art bearingless slice motors with a reluctance rotor [10] and a hysteresis rotor [13], respectively. This characteristic can be attributed to the Halbach array that can be modeled as a flux source in parallel with a relatively large source reluctance. The large source reluctance can reduce the sensitivity of the air-gap flux to the rotor's position, thereby reducing the negative stiffness. The stiffness values from FEA simulations are listed in Table III.

TABLE III
SUSPENSION PERFORMANCE PARAMETERS

Symbol	Quantity	Value
k_θ	Tilting stiffness	34.4 mNm/deg
k_r	Radial stiffness	-26.6 N/mm
k_z	Axial stiffness	15.3 N/mm
$ k_r/k_z $	Stiffness ratio	1.7
K_f	Force constant	3.5 N/A

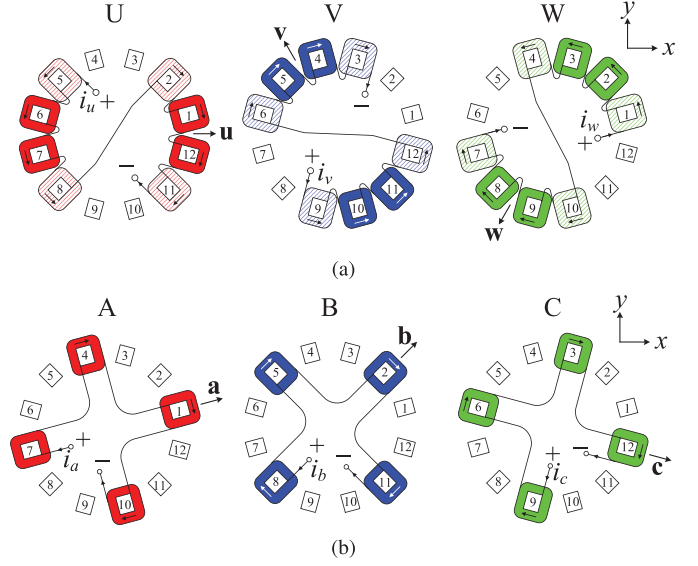


Fig. 7. Winding scheme. (a) Suspension winding consisting of 12 N_1 -turn coils (solid color) and 12 N_2 -turn coils (shaded color) connected in three phases U, V, and W. (b) Rotation winding consisting of 12 N_r -turn coils connected in three phases A, B, and C.

C. Winding

The stator winding consists of 36 coils made of a bondable magnet wire (21 AWG) in three different numbers of turns. Their vertical arrangement is shown in Fig. 2 and horizontal arrangement is shown in Fig. 7. The coils are interconnected to form two separate three-phase windings: suspension windings and rotation windings.

1) Suspension Winding: The detailed coil connection in the suspension winding is shown in Fig. 7(a). Here, the three phases U, V, and W are color coded with red, blue, and green, where the solid color denotes the N_1 -turn coils and the shaded color denotes the N_2 -turn coils. Each phase consists of four N_1 -turn and four N_2 -turn coils distributed over the stator teeth. The coils in the same phase are connected in series, and therefore conduct the same current i_u , i_v , and i_w , respectively. The three phases are staggered such that their magnetic axes $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are apart from each other by 120° mechanically. Excited by a set of balanced three-phase currents

$$\begin{aligned} i_u &= I_s \cos(\phi_s) \\ i_v &= I_s \cos(\phi_s - 2\pi/3) \\ i_w &= I_s \cos(\phi_s + 2\pi/3) \end{aligned} \quad (2)$$

the suspension winding generates a two-pole MMF whose magnitude and angle are controlled by I_s and ϕ_s , respectively.

We have optimized the turn ratio N_2/N_1 to make the two-pole MMF as sinusoidal as possible, thereby minimizing the suspension force interference between the x -axis and y -axis. Referring to Fig. 7(a), we can express the suspension MMF on the n th stator tooth as

$$\mathcal{F}_s[n] = \sum_m \mathbf{N}_s[n, m] \mathbf{i}_s[m] \quad (3)$$

where $\mathbf{N}_s$ is the coil arrangement matrix and $\mathbf{i}_s$ is the current vector defined as

$$\mathbf{N}_s = \begin{bmatrix} N_1 & 0 & -N_2 \\ N_2 & 0 & -N_1 \\ 0 & N_2 & -N_1 \\ 0 & N_1 & -N_2 \\ -N_2 & N_1 & 0 \\ -N_1 & N_2 & 0 \\ -N_1 & 0 & N_2 \\ -N_2 & 0 & N_1 \\ 0 & -N_2 & N_1 \\ 0 & -N_1 & N_2 \\ N_2 & -N_1 & 0 \\ N_1 & -N_2 & 0 \end{bmatrix}, \quad \mathbf{i}_s = \begin{bmatrix} i_u \\ i_v \\ i_w \end{bmatrix}. \quad (4)$$

Here $n \in \{1, \dots, 12\}$ is the stator tooth index and $m \in \{1, 2, 3\}$ is the phase-current index. The turn numbers with negative signs are of those coils in the opposite polarity. Now, let $\tilde{\mathcal{F}}_s[k]$ be the discrete Fourier transform of $\mathcal{F}_s[n]$, where $k \in \{1, \dots, 12\}$ is the spatial frequency index. This allows us to define the total harmonic distortion of $\mathcal{F}_s[n]$ as

$$\text{THD}(\mathcal{F}_s[n]) = \sqrt{\frac{\sum_{k \neq 1} |\tilde{\mathcal{F}}_s[k]|^2}{|\tilde{\mathcal{F}}_s[1]|^2}}. \quad (5)$$

Our objective is to find the turn ratio N_2/N_1 that minimizes the THD of $\mathcal{F}_s[n]$, i.e.,

$$\min_{N_2/N_1} \text{THD}(\mathcal{F}_s[n]) \quad (6)$$

subject to the constraint in (2). The optimization problem is numerically solved, and the minimum THD ≈ 0 is achieved at $N_2/N_1 = 0.366$. We observe that the optimality remains the same regardless of the I_s and ϕ_s . This optimal turn ratio agrees with the value reported in [21], which accounts for continuous-space Fourier series up to the 19th harmonics to reduce cross-axis interference forces.

The effect of the optimal coil design on suspension force generation is tested via 3-D magnetostatic FEA simulations. In the simulation, the two-pole MMF direction is set to $\phi_s = 0$ so that the suspension force is nominally toward the x -axis. Two winding designs are simulated: one with the optimal turn ratio $N_2/N_1 = 0.366$ and the other with a nonoptimal turn ratio $N_2/N_1 = 1$. In both cases, the sum of the MMFs through the two coils, i.e., $(N_1 + N_2)I_s$, is varied from 0 to 4000 At. This excitation condition ensures a fair comparison in terms of

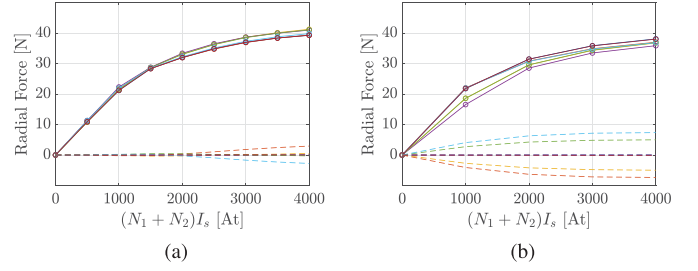


Fig. 8. Computed radial forces versus suspension winding MMF: (a) for the optimal turn ratio $N_2/N_1 = 0.366$ and (b) for a nonoptimal turn ratio $N_2/N_1 = 1$. The solid lines show the x -axis forces and the dashed lines show y -axis forces, where the curves are parametric on the rotor mechanical angle ϕ .

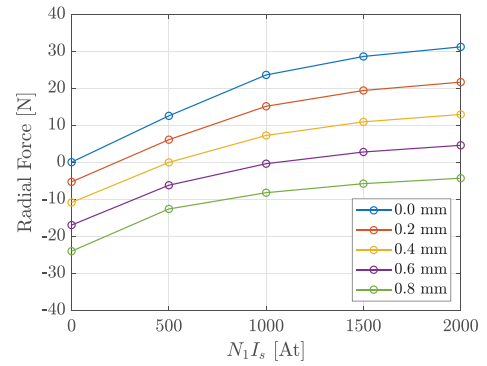


Fig. 9. Computed radial forces versus suspension winding MMF. The curves are parametric on the rotor radial offset δr .

the winding current density. The rotor mechanical angle ϕ , defined in Fig. 3, is varied between 0° and 90° in 15° increments. Fig. 8(a) shows the computed radial forces when the turn ratio is optimal, i.e., $N_2/N_1 = 0.366$. Here, the solid lines show the x -components and the dashed lines show the y -components, where the curves are parametric on ϕ . As the MMF increases, the x -axis forces increase, whereas the y -axis forces remain approximately zero up to 2000 At. This means that cross-axis force interference is ignorable for all ϕ especially in a linear range. On the other hand, we observe a considerable amount of force interference from a nonoptimal coil design, e.g., $N_2/N_1 = 1$, as shown in Fig. 8(b). Here, the MMF generates not only the x -axis forces, but also y -axis forces whose amplitudes fluctuate with the rotor mechanical angle ϕ . Our actual coils are fabricated with $N_1 = 140$ and $N_2 = 52$, which leads to $N_2/N_1 \approx 0.37$.

We have tested the takeoff capability of the suspension winding in 3-D magnetostatic FEA simulations. In the simulation, the two-pole MMF is generated toward the positive x -axis ($\phi_s = 0$), while the rotor is displaced from the stator center by δr to the negative x -axis. Then, the MMF of the primary coil $N_1 I_s$ is increased from 0 to 2000 At. Fig. 9 shows the resulting radial forces parametric on δr . The simulation results predict that the suspension winding can lift off the rotor for an offset $\delta r \leq 0.6$ mm with a current $I_s < 7.14$ A. The estimate of the force-current sensitivity is $K_f = 3.5$ N/A.

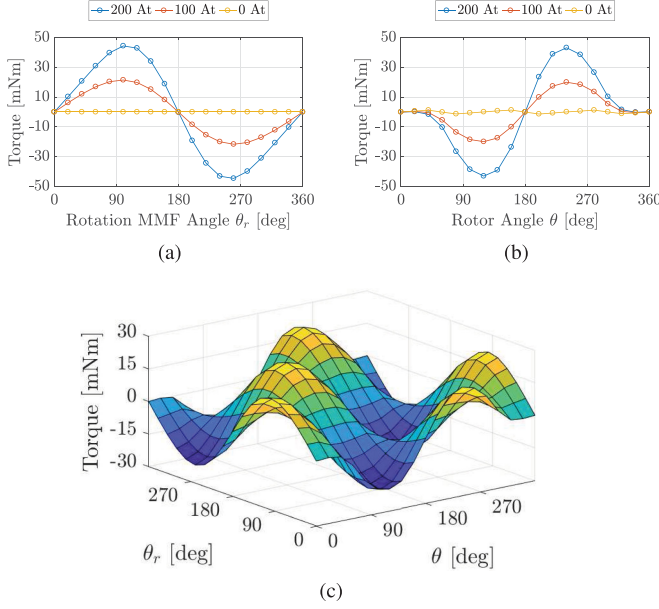


Fig. 10. Computed torques from FEA simulations: (a) with respect to MMF angle θ_r in electric degrees, (b) with respect to rotor angle θ in electric degrees, and (c) with respect to θ_r and θ . The curves in (a) and (b) are parametric on the MMF magnitude $\mathcal{F}_r$, whereas the surface in (c) is for $\mathcal{F}_r = 100$ At.

2) Rotation Winding: The detailed coil connection in the rotation winding is shown in Fig. 7(b). Here, the three phases A, B, and C are color coded with red, blue, and green. The three phases are staggered such that their magnetic axis **a**, **b**, and **c** are apart from each other by 30° mechanically. Excited by a set of balanced three-phase currents

$$\begin{aligned} i_a &= I_r \cos(p\phi_r) \\ i_b &= I_r \cos(p\phi_r - 2\pi/3) \\ i_c &= I_r \cos(p\phi_r + 2\pi/3) \end{aligned} \quad (7)$$

where $p = 4$ is the number of pole pairs, the rotation winding generates an eight-pole MMF whose magnitude and angle are controlled by I_r and ϕ_r , respectively. This generates a torque by interacting with the eight-pole rotor flux.

We have evaluated the torque characteristic in 3-D magnetostatic FEA simulations. The simulation parameters are the rotation MMF magnitude $\mathcal{F}_r \equiv N_r I_r$, rotation MMF electrical angle $\theta_r \equiv p\phi_r$, and rotor electrical angle $\theta \equiv p\phi$. Fig. 10(a) shows the computed torques from a case where we fix the rotor angle to $\theta = 0^\circ$ and vary the rotation MMF angle θ_r from 0° to 360° . The curves are parametric on $\mathcal{F}_r$. We observe that the torque is approximately sinusoidal with respect to θ_r and linear in $\mathcal{F}_r$, which resembles synchronous machines.

Fig. 10(b) shows the computed torques from another case where we fix the MMF angle to $\theta_r = 0^\circ$ and vary the rotor angle θ from 0° to 360° . Note that the curve for $\mathcal{F}_r = 0$ At shows cogging torque, which is small in amplitude (2.6 mNm) due to the rotor teeth design discussed in Section III-B. The torque curves for $\mathcal{F}_r = 100$ At and $\mathcal{F}_r = 200$ At appear to

TABLE IV
ROTATION PERFORMANCE PARAMETERS

Symbol	Quantity	Value
J_o	Rated current density	$3 \text{ A}_{\text{rms}}/\text{mm}^2$
V_B	Amplifier bus voltage	$\pm 60 \text{ V}$
K_τ	Torque constant	15.4 mNm/A
τ_o	Rated torque	26.8 mNm
Ω_o	Rated speed	3486 r/min
P_o	Rated power	9.8 W
η_o	Rated power efficiency	64%

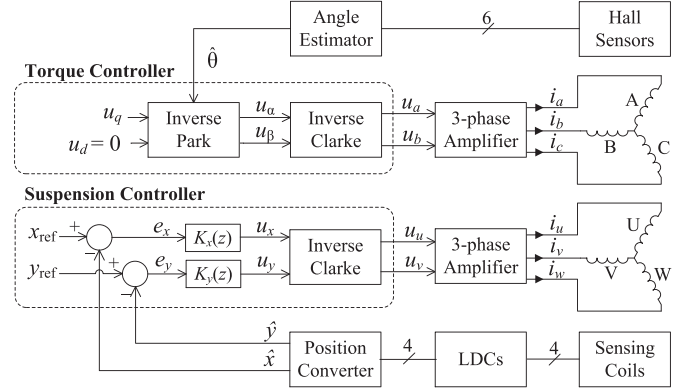


Fig. 11. Control system diagram. Two separate control algorithms are implemented for suspension control and rotation control.

have second-order distortions. That is, the peak torques do not occur exactly at $\theta = \pm 90^\circ$. This can be attributed to the effect of reluctance torque due to the rotor saliency [22, p. 290].

Fig. 10(c) shows a surface plot of the computed torques. Here, the torques are computed for $\mathcal{F}_r = 100$ At, and both of the MMF angle θ_r and the rotor angle θ are swept from 0° to 360° . The plot shows that the peaks of the torque surface are approximately aligned over a line

$$\theta_r = \theta + 90^\circ. \quad (8)$$

The computed torques over this line has an average of 14.3 mNm and an amplitude ripple of 7.4 mNm. Such a torque ripple is attributable to the effect of stator saliency, which can be reduced with a fine-tooth stator core [17].

The actual rotation coils are fabricated for $N_r = 108$ turns, which leads to a torque constant prediction of $K_\tau = 15.4 \text{ mNm/A}$. For a current density $J_o = 3 \text{ A}_{\text{rms}}/\text{mm}^2$ and bus voltage $V_B = \pm 60 \text{ V}$, we calculate that the rated torque, speed, and power of the motor are $\tau_o = 26.8 \text{ mNm}$, $\Omega_o = 3486 \text{ r/min}$, and $P_o = 9.8 \text{ W}$, respectively. The rated power efficiency is estimated to $\eta_o = 64\%$ ignoring hysteresis and eddy-current losses. The motor parameters are listed in Table IV.

IV. CONTROL SYSTEM

The prototype bearingless motor is used in a closed-loop system with sensors, power amplifiers, and controllers, as shown in Fig. 11. This section explains the implementation details of the control system.

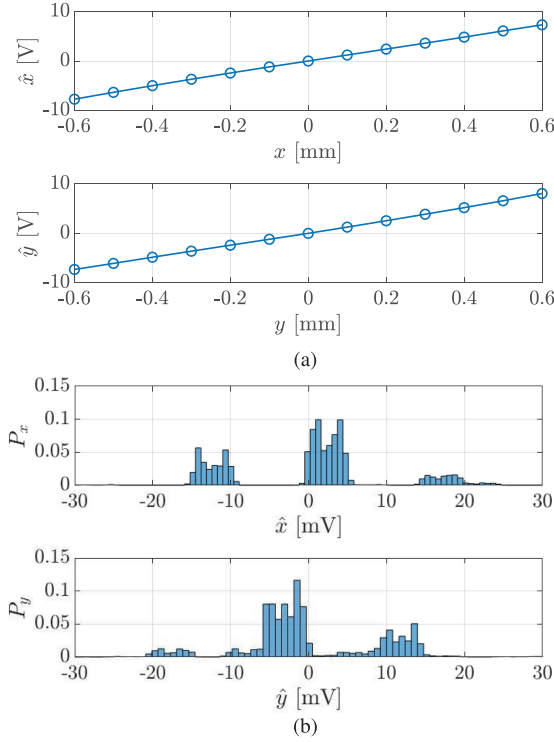


Fig. 12. Eddy-current position sensor outputs $\hat{x}$ and $\hat{y}$. (a) Sensor outputs versus rotor positions. (b) Noise histograms.

A. Sensors

As shown in Fig. 3, the motor stator is equipped with four eddy-current based displacement sensors, c_1 – c_4 , and six Hall-effect sensors, h_1 – h_6 . The eddy-current sensors measure the distances to the rotor inner surface, from which we compute the rotor radial positions $\hat{x}$ and $\hat{y}$. The Hall sensors measure the leakage bias flux density between the stator teeth, based on which we estimate the rotor electrical angle $\hat{\theta}$.

1) Eddy-Current Sensors: We have designed and implemented low-cost digital eddy-current sensors, as presented in [23]. The sensor mainly consists of a sensing coil (WR111180-36F5-B1, TDK Corporation), capacitor, and inductance-to-digital converter (LDC1101, Texas Instruments). The sensing coil and capacitor form an oscillator whose resonance frequency varies with the coil-rotor distance due to the eddy currents. The LDC measures resulting frequency changes. That is, the LDC drives the oscillator at the self-resonance, counts the number of oscillations over a time window, and outputs an instantaneous resonance frequency. The time window is set to 500 μ s, which leads to a data rate of 2 kHz. To avoid cross-axis coupling, the operating frequencies of the four oscillators are stagger-tuned (2.9–3.9 MHz) with different capacitance values (100–180 pF).

The rotor positions $\hat{x}$ and $\hat{y}$ are computed by taking differential outputs of the diametrically opposing sensors, i.e., $c_1 - c_3$ and $c_2 - c_4$, followed by scaling and analog conversion. Fig. 12 shows the measured sensor outputs $\hat{x}$ and $\hat{y}$. As shown in Fig. 12(a), the sensor outputs are approximately linear in ± 0.6 mm range with the x -axis and y -axis sensitivities of 12.0

and 12.3 V/mm, respectively. For this measurement, the eddy-current sensing module is mounted on a structure fixed on an optical breadboard, and the rotor is displaced with respect to the sensing module using a two-axis micrometer stage. Fig. 12(b) shows that the measured standard deviations, i.e., sensor resolutions, are $\sigma_{\hat{x}} = 9.6$ mV (0.80 μ m) and $\sigma_{\hat{y}} = 8.8$ mV (0.72 μ m). For this measurement, the rotor is mechanically fixed with respect to the eddy-current sensing module and the sensor outputs are acquired with a real-time controller (NI cRIO-9064) equipped with an analog input module (NI 9205) that has a 350-kHz bandwidth, 16-bit resolution, and ± 10 V input range. The data acquisition rate is synchronized to the sensing module's sampling rate, i.e., $f_s = 2$ kHz. A total of 1.2×10^5 data points are collected over 60 s to compute the noise statistics.

2) Hall Sensors: The motor stator has six Hall sensors (SS49E, Honeywell) inserted between the teeth. Each sensor measures the radial component of the fringe field, which varies with the rotor angle θ as well as the radial displacement. We take sums of the outputs of the diametrically opposing sensors, i.e., $e_1 = h_1 + h_4$, $e_2 = h_2 + h_5$, and $e_3 = h_3 + h_6$, to average out the effect of two-pole flux in the air gap, such as those induced by the rotor radial displacement.

The signals e_1 , e_2 , and e_3 are periodic with respect to the rotor angle θ , and they can be approximately modeled as three-phase sinusoidal signals

$$\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} \approx \begin{bmatrix} \cos(\theta + \pi/3) \\ \cos(\theta - \pi/3) \\ \cos(\theta - \pi) \end{bmatrix} = \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ 1/2 & \sqrt{3}/2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}. \quad (9)$$

By defining $\mathbf{y} \equiv \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}^\top$ as the measurement and $\mathbf{x} \equiv [\cos \theta \ \sin \theta]^\top$ as the estimand, we can formulate a linear estimation problem $\mathbf{y} \approx H\mathbf{x}$ whose minimum mean square error estimate $\hat{\mathbf{x}} \equiv [\widehat{\cos \theta} \ \widehat{\sin \theta}]^\top$ is $(H^\top H)^{-1} H^\top \mathbf{y}$. Then, the rotor angle estimate is computed as

$$\hat{\theta} = \arctan \left(\frac{\widehat{\sin \theta}}{\widehat{\cos \theta}} \right). \quad (10)$$

The angle estimate $\hat{\theta}$ is used to implement motor commutation and torque control.

B. Power Amplifiers

We employ two linear three-phase transconductance amplifiers (LA-210 T, Varedan) to separately drive the suspension windings and rotation windings.

The suspension amplifier takes current references u_u and u_v and drives the currents through the suspension winding such that $i_u = g_s u_u$, $i_v = g_s u_v$, and $i_u + i_v + i_w = 0$. Here, g_s is the transconductance whose dc gain and bandwidth are set to 1 A/V and 5.5 kHz, respectively. The chosen amplifier can drive a peak current up to 12 A, satisfying the current requirement for the rotor takeoff from 0.6 mm offset.

The rotation amplifier takes current references u_a and u_b and drives the currents through the rotation winding such that

$i_a = g_r u_a$, $i_b = g_r u_b$, and $i_a + i_b + i_c = 0$. The dc gain and bandwidth of g_r are set to 1A/V and 906 Hz, respectively. The chosen amplifier can supply a terminal voltage up to ± 60 V and drive a maximum rms current of $2.5 A_{\text{rms}}$.

C. Control Algorithm

As shown in Fig. 11, the control algorithm consists of two decoupled controllers: suspension controller and torque controller. These are implemented in a field-programmable gate array (FPGA) of a real-time controller (NI cRIO-9064).

1) Suspension Controller: The rotor radial displacements x and y are open-loop unstable. The suspension controller stabilizes these two via feedback control. It takes the position measurements $\hat{x}$ and $\hat{y}$, computes stabilizing control signals u_u and u_v , and sends them to a transconductance amplifier.

The level of radial instability can be measured with an unstable pole frequency $\omega_u = \sqrt{|k_r/m|}$, where k_r is the negative radial stiffness and m is the rotor mass. With $k_r = -26.6$ N/mm and $m = 125$ g, the unstable pole of our plant is at about $\omega_u = 73$ Hz. To stabilize the radial rotor position, we implement two single input single output controllers $K_x(z)$ and $K_y(z)$ for the x -axis and y -axis separately. Specifically, we implement discrete-time lead controllers whose transfer functions are

$$K_x(z) = K_y(z) = k_p \frac{z + b_0}{z + a_0}. \quad (11)$$

The values $b_0 = -0.8019$ and $a_0 = 0.04742$ are obtained by taking the Tustin transformation of a continuous-time lead compensator that has a zero frequency at 70 Hz and a pole frequency at 700 Hz. Such a zero frequency is selected to add phase lead above the unstable pole frequency ω_u , with an aim of shaping the loop return ratio $L(j\omega)$ to cross the unity magnitude at a frequency $\omega_c > \omega_u$ with a positive phase margin. The pole frequency is selected for a typical pole-zero frequency ratio $\alpha = 10$. The proportional gain is set to $k_p = 2.5$. The transfer function (11) is implemented in synchronism with the data rate of the eddy-current sensor, i.e., $f_s = 2$ kHz.

The outputs u_x and u_y from the controllers $K_x(z)$ and $K_y(z)$ are transformed to the control signals u_u and u_v via

$$\begin{bmatrix} u_u \\ u_v \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1/2 & \sqrt{3}/2 \end{bmatrix} \begin{bmatrix} u_x \\ u_y \end{bmatrix}. \quad (12)$$

The transformation (12), often referred to as the Inverse Clark Transformation, performs a coordinate transformation from the xy -frame to the three-phase uvw -frame. The discrete-time signals u_u and u_v are upsampled by a factor of 50 in a faster loop (100 kHz), filtered through a 32-point moving average filter, and sent out to the suspension amplifier. The upsampling and averaging helps the linear amplifier avoid voltage saturation due to reducing the size of update steps in the analog output.

2) Torque Controller: Our goal in torque control is to commute the three-phase currents (7) such that the rotation MMF is always in quadrature ahead of the rotor flux (8). To this end, we adopt a scheme called field-oriented control [22]. This allows us to design control signals u_d and u_q that individually control the in-phase and quadrature components of the MMF

with respect to the rotor flux. For our purpose, we set $u_d = 0$ and use u_q to control the torque.

The control signals u_d and u_q are transformed via a series of transformation matrices

$$\begin{bmatrix} u_a \\ u_b \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1/2 & \sqrt{3}/2 \end{bmatrix} \begin{bmatrix} \cos \hat{\theta} & -\sin \hat{\theta} \\ \sin \hat{\theta} & \cos \hat{\theta} \end{bmatrix} \begin{bmatrix} u_d \\ u_q \end{bmatrix} \quad (13)$$

to compute the current references u_a and u_b for the transconductance amplifier. The first matrix, often referred to as the Inverse Park Transformation, performs a rotational transformation on $[u_d \ u_q]^T$ using the angle estimate $\hat{\theta}$ in (10) to compute intermediate variables $[u_\alpha \ u_\beta]^T$. The second matrix performs a coordinate transformation on $[u_\alpha \ u_\beta]^T$ to compute $[u_a \ u_b]^T$.

The torque control algorithm in (13) is implemented at a sampling rate of 10 kHz. The current references u_a and u_b are then upsampled by a factor of 10 in a faster loop (100 kHz), filtered through an 8-point moving average filter, and sent out to the rotation amplifier as a smoothed waveform.

V. TEST RESULTS

The prototype system can levitate the rotor and drive it up to 5500 r/min in air. This section discusses the test setup and results in detail.

A. Suspension Test

The magnetic gap between the stator wall and the rotor outer surface is $g_o = 2$ mm when the rotor and stator are concentric. This large gap allows the use of a fluid-tight plastic pump housing to enclose the rotor. Plastic shims are inserted in this gap so that the maximum reachable rotor off-centering is about $\delta r = 0.5$ mm. As we increase the proportional gain in (11) from $k_p = 0$ to $k_p = 2.5$, the rotor takes off from the wall and levitates at the stator center. For a smooth takeoff, the step change in k_p is filtered by a first-order low-pass filter that has a time constant of 1 s, although the takeoff is still possible without the low-pass filter.

We have measured the response of $\hat{x}$ to a small-signal input step x_{ref} whose peak-to-peak amplitude is $33 \mu\text{m}$ (0.04 V). The measured step response, shown in Fig. 13(a), exhibits an overshoot of about 55% and rise time (10–90%) of about $t_r = 3$ ms. We have also measured the loop return ratio $L(j\omega)$ of the suspension system using a digital dynamic signal analyzer implemented in the real-time processor (cRIO-9064). The Bode plot in Fig. 13(b) shows the magnitude $|L(j\omega)|$ and phase $\angle L(j\omega)$ curves, where the measured cross-over frequency is $\omega_c = 84$ Hz with a phase margin of $\phi_m = 11.3^\circ$. The relatively small phase margin can be attributed to the overall loop delay arising from multiple sources, e.g., relatively slow loop rate (2 kHz), latency in the digital eddy-current sensors, moving-average filtering on the controller outputs, and eddy currents induced on the solid-steel rotor. Although the phase margin is relatively small, we have not observed suspension failures during the rotation tests, and therefore deduce that the suspension control is robust enough to withstand the force disturbances generated from the rotation.

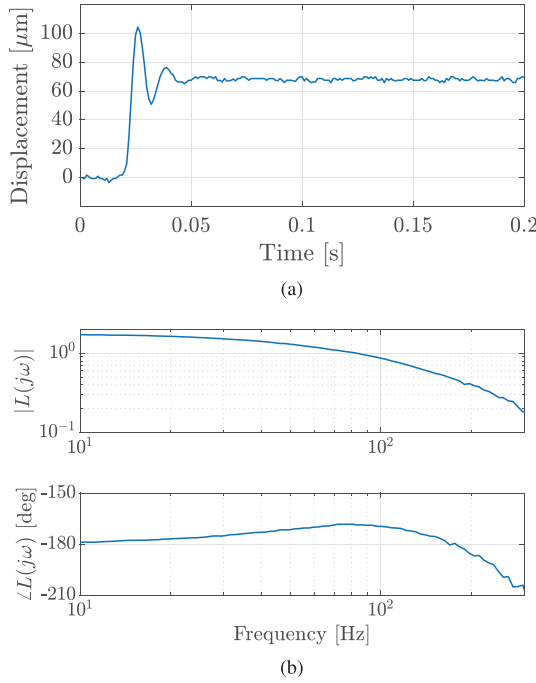


Fig. 13. Suspension test results. (a) Step response. (b) Frequency response of the loop return ratio.

B. Rotation Test

Once the rotor is levitated at the stator center, we can rotate the rotor by assigning a nonzero value to the q -axis current command u_q . The rotation test is performed in two modes: steady-state and transient modes.

In the steady-state test, we directly assign a fixed value to u_q and let the rotor reach a steady-state speed. Fig. 14(a) shows the relation between u_q and the steady-state speed measured with a digital tachometer (DT-2231C⁺). The steady state occurs due to the balance between the motor torque and the air drag. As u_q is increased from 0 to 1.5 V, which maps to the q -axis current i_q from 0 to 1.5 A within the amplifier's bandwidth, the steady-state rotor speed gradually increases and plateaus at around 5500 r/min. The speed saturation is due to the limited power bandwidth of the amplifier [2, p.129] caused by the relatively low-bus voltage (± 60 V) and large phase-to-phase inductance (37 mH) of the rotation winding. Indeed, we observe that the winding terminal voltage is clamped to the bus voltage when the rotational speed is above 5200 r/min.

During the transient test, u_q is switched between two constant values so as to evaluate the rotor acceleration under a constant driving torque. The switching moments of u_q are determined by a hysteretic controller whose lower and upper thresholds are set to 500 and 1500 r/min. The rotor speed is measured with a speedometer implemented in an FPGA, which computes the speed by measuring the elapsed time between the zero-crossing points of $\hat{\theta} \in (-\pi, \pi)$. Fig. 14(b) shows the measured transient speeds. Here u_q is switched between ± 0.2 V in the top trace and ± 1 V in the bottom trace. The resulting waveforms are triangular, indicating that the rotor experiences approximately

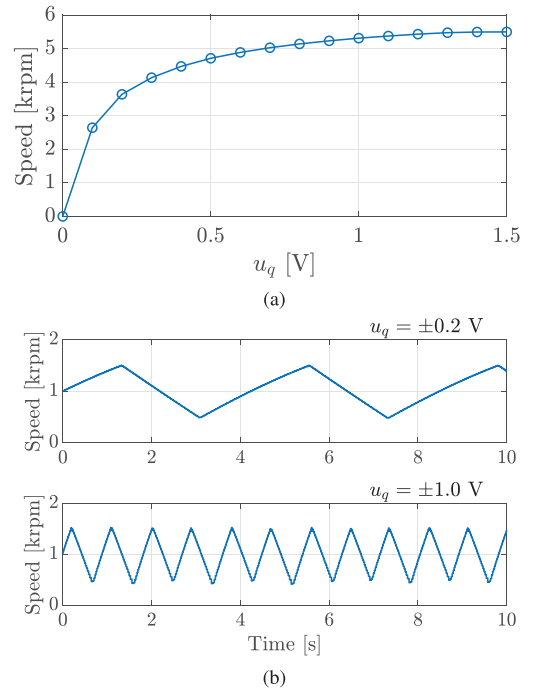


Fig. 14. Rotation test results. (a) Steady-state speeds as a function of drive amplitudes. (b) Transient speed responses.

constant accelerations. We can infer the motor torque τ using Newton's second law $\tau = I_{zz} \frac{d\omega_z}{dt}$, where ω_z is the rotor speed and $I_{zz} = 51.6 \text{ kg} \cdot \text{mm}^2$ is the rotor's moment of inertia. In the bottom trace of Fig. 14(b), we estimate the torque $\hat{\tau} = 14.9 \text{ mNm}$ from a rising edge and $\hat{\tau} = 13.4 \text{ mNm}$ from a falling edge, and thus the respective torque constants $\hat{K}_\tau = 14.9 \text{ mNm/A}$ and $\hat{K}_\tau = 13.4 \text{ mNm/A}$, given that the amplifier's transconductance is 1 A/V. The measured torque constants are smaller (3% and 13%, respectively) than the torque constant prediction $K_\tau = 15.4 \text{ mNm/A}$ from FEA simulations in Section III-C2. We hypothesize that such a discrepancy is related to dynamic effects, e.g., windage, hysteresis, and eddy-current losses.

VI. CONCLUSION

In this article, we presented a new type of homopolar bearingless slice motor that levitates and rotates a solid-steel reluctance rotor. Our magnetic design combines the force generation mechanism of flux-biased magnetic bearings and the torque generation mechanism of synchronous motors, in such a way that the two mechanisms do not interfere with each other. This design enables decoupled control of suspension and rotation. The homopolar flux biasing was implemented in a novel fashion with Halbach magnet arrays. This enables a relatively large air-gap machine (2 mm), whose torque and speed performances are acceptable for use in pumps, such as an extracorporeal blood pump. Our bearingless motor is particularly favorable for such an application because the air gap is large enough to accommodate various inserts, such as rotor encapsulation, fluid flow channel, and the pump housing, and the magnet-free rotor can reduce the repeated costs of the disposable pump modules.

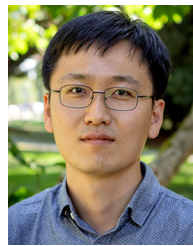
Although satisfactory as our phase I effort, this article has room for further improvements. For example, the FEA simulation in Section IV-B predicts significant torque ripples. To resolve this issue, we might redesign the rotor with sinusoidal teeth and the stator with fine teeth, so as to reduce the higher harmonics of the air-gap flux distribution. The rotation and suspension characteristics of the prototype system were evaluated via 3-D magnetostatic FEA, assuming the effects of eddy currents and hysteresis are ignorable. We plan to take the dynamic effects into account in the future study. Also, the present prototype has the position sensors occupying space inside the rotor. Such a sensor arrangement complicates the pump module design because an inner rotor wall must be accessible from outside the housing. A better design would be relocating the sensors to the outside of the rotor. This can be potentially implemented with a redundant number of Hall sensors inserted between the stator teeth, based on which an estimator computes the rotor's transnational and rotational positions simultaneously.

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